European Journal for Philosophy of Science (2026) 16:12
https://doi.org/10.1007/s13194-025-00714-9

PAPER IN THE PHILOSOPHY OF THE LIFE SCIENCES

Rethinking interlevel experiments: no remainder from evidence for causal relations

Maria Şerban[1]

Received: 2 August 2024 / Accepted: 22 December 2025 / Published online: 5 February 2026

Abstract

This paper examines the transformation of Craver's (2009) mutual manipulability (MM) account into the matched interlevel experiments (MIE) framework (Craver et al., 2021) and argues that it amounts to a theoretical reduction of mechanistic constitutive relations to causal mediation. While the MIE account successfully resolves the incoherence challenge that plagued MM, it does so by eliminating the distinctive theoretical content that constitutive categories were supposed to provide. The processual reframing that enables this solution replaces hierarchical part-whole relationships with temporal causal sequences, changing what mechanistic explanations are understood to accomplish. Drawing on paradigmatic action potential experiments, I demonstrate that practices satisfying MIE's formal requirements consistently establish causal mediation relationships without requiring constitutive interpretation. I address several theoretical defenses of constitutive categories—including interpretive objections about two types of constitution, arguments for distinctive explanatory value, and appeals to mechanistic levels—showing that none can rescue constitutive distinctiveness once constitution is explicitly identified with causal betweenness. Rather than undermining mechanistic approaches, this analysis suggests that their explanatory power derives from methodological sophistication in investigating complex, multi-scale causal structures rather than from categorically distinct constitutive relationships.

Keywords Mechanisms · Constitutions · Causal mediation · Action potential · Interlevel experiments

✉ Maria Şerban
m.serban@uea.ac.uk; mariaserban.org@gmail.com

[1] Department of Philosophy, University of East Anglia School of Politics, Philosophy, and Area Studies , Norwich Research Park, AHB 0.21, Norwich NR4 7TJ, UK

1 Introduction

Substantive research efforts in neuroscience are invested in discovering how complex physical systems composed of intricately organised and interacting parts bring about a diverse range of biological and psychological phenomena ranging from nerve excitation and synaptic plasticity to spatial memory and decision making. Reflecting on such scientific practices, philosophers have attempted to systematise these efforts under the umbrella of the new mechanistic philosophy (e.g., Bechtel & Richardson, 1993/2010; Darden, 1991; Craver, 2007a, b, c). Central to this philosophical program has been the claim that genuine mechanistic explanations are "more than" merely causal explanations—they provide information not only about which factors causally trigger a phenomenon, but also about what components are constitutively relevant as parts of the hypothesised mechanism.

The most influential defence of this causation-constitution distinction has come from Craver and his collaborators (Craver & Bechtel, 2007; Craver, 2007a, b, c; Kaplan & Craver, 2011; Craver & Darden, 2013)[1]. A key feature of their approach lies in its methodological commitment to bridge philosophical accounts with actual neuroscientific practice, developing frameworks that not only characterise what mechanistic explanations are, but also specify the experimental strategies through which mechanistic knowledge is discovered and validated. While this continuous engagement with experimental practice has led to many refinements, the most recent version of this philosophical project (Craver et al., 2021) makes it unclear whether any substantive distinction between the mechanistic categories of causation-constitution is still to be had.

The original mutual manipulability (abbreviated MM) account (Craver, 2007a, b, c; Kaplan & Craver, 2011) argued that constitutive dependence—the part-whole relationship between proper parts and the mechanisms they compose—can be established through coordinated pairs of bottom-up and top-down interlevel experiments, analysed within a Woodwardian interventionist framework (Woodward, 2003). This approach suggested that experimental practices could deliver evidence not merely about causal relationships, but also about the constitutive structure of biological mechanisms themselves.

However, the most recent version of this project—the matched interlevel experiments (MIE) account (Craver et al., 2021)—was developed partly in response to various criticisms and partly to better align with interventionist principles. The account introduces a puzzling theoretical development: constitutive relevance relations are now characterised in terms of causal mediation. This reformulation generates a multifaceted challenge. If constitutive relations are essentially a form of causal mediation, what distinctive work does the concept of constitution perform in mechanistic explanation? How should we understand the relationship between discovering causal relations and discovering constitutive relations in neuroscientific practice? And what

[1] For defences of the distinction between mechanistic constitution and mechanistic causation that leverage more metaphysical arguments, some influential accounts include: Glennan, 2002, 2017; Krickel, 2018a, b; Kaiser & Krickel, 2017; Gillett, 2020.

implications does this have for how we characterise the logic and aims of mechanistic research in neuroscience?

This paper addresses these questions by examining the transformation of the MM into the MIE account and evaluating the latter's relationship to experimental practice in neuroscience. I argue that while MIE provides valuable resources for understanding what Craver, Glennan and Povich call *interlevel* experiments, the attempt to justify constitutive relations in terms of these experimental practices faces significant challenges. The proposed analysis suggests that the distinctive features often attributed to mechanistic explanation—particularly those involving constitutive as opposed to causal relationships—may be better understood in terms of pragmatic and modal considerations rather than as marking a fundamental theoretical distinction.

The paper proceeds as follows. Section 2 traces the theoretical development from the MM to the MIE account, examining the motivations for this transition and its conceptual implications. Section 3 analyses whether experimental practices in neuroscience support the causal mediation account of constitution, articulating the logic of interlevel experiments and their relationship to the MIE framework through concrete case studies from action potential research. Section 4 unpacks what I term the theoretical redundancy argument, examining whether mechanistic constitutive claims provide distinctive explanatory resources beyond those available through causal analysis. I first discuss an interpretive objection (whether the reduction reflects mere terminological clarification about two types of constitution), and then evaluate one sophisticated philosophical defence of the explanatory value of mechanistic constitution. Section 5 addresses a final structural objection: whether mechanistic levels—either understood traditionally as hierarchical compositional structure or more recently as pragmatically useful heuristic tools—can vindicate constitutive categories even if individual constitutive relations reduce to causal mediation. The conclusion summarises the argument and explores the prospects for developing alternative accounts of what makes mechanistic explanations distinctive.

2 From mutual manipulability to matched interlevel experiments

Craver's original analysis of mechanistic constitutive relevance (Craver, 2007a, b, c) was rooted in an epistemological question: What could count as sufficient evidence, in practice, to establish when an entity is a constitutive component of a biological mechanism? In Craver's work, the question was motivated by both philosophical and empirical considerations about what makes mechanistic explanations in fields like neuroscience distinctive.

Philosophically, mechanisms have been conceived as hierarchically organised part-whole systems, where components stand in constitutive rather than causal relationships to the phenomena they produce (e.g., Bechtel & Richardson, 1993/2010; Machamer et al., 2000). In line with this conception, mechanistic explanations were said to reveal "how things work" by showing how organised parts give rise to higher-level phenomena—they are explanatory precisely because they describe the structures that produce, maintain, or underlie the explanandum phenomenon, rather than merely describing regularities or correlations (Craver, 2007a, b, c; Kaplan & Craver, 2011).

Empirically, the distinction between constitutive relevance and causal relevance claims seemed supported by specific experimental practices used in neuroscience. While standard two variables causal experiments investigate relationships between distinct entities, so-called *interlevel* experiments appear to probe constitutive relationships *within* mechanisms. The joint motivation, then, was to articulate a view of mechanistic explanations as being more than "merely" causal etiological explanations. On this view, their epistemic value resides in helping us understand both the causal relationships that obtain between distinct entities and the constitutive relationships that obtain between mechanistic parts and wholes (Fagan, 2015).

To address this challenge, Craver proposed that constitution could be experimentally demonstrated when both part-whole relationship and part-whole mutual manipulability conditions are satisfied[2]. According to the original MM account, the following conditions are *sufficient* to establish that an activity φ of entity X is constitutively relevant to the mechanism for S's ψ-ing:

"(i) X is part of S; (ii) in the conditions relevant to the request for explanation, there is some change to X's φ-ing that changes S's ψ-ing, and (iii) in the conditions relevant to the request for explanation, there is some change to S's ψ-ing that changes X's φ-ing." (Craver, 2007a, b, c: 153).

Building on Woodward's (2003) manipulationist counterfactuals account, conditions (ii) and (iii) have been further specified as:

(CR1): when φ is set to the value φ_1 in an ideal intervention I_1, then $\psi = f(\varphi_1)$, and.

(CR2): when ψ is set to the value ψ_1 in an ideal intervention I_2, then $\varphi = g(\psi_1)$.

Furthermore, Craver mapped these formal conditions onto three types of interlevel experiments[3] commonly used in neuroscience: bottom-up inhibitory experiments (involving deletion or inhibition of a component), bottom-up excitatory experiments (involving stimulation of a component), and top-down activation experiments (involving activation of the whole mechanism while detecting changes in parts). The key insight was that these experimental practices, when jointly satisfied, could demonstrate that a component stands in a constitutive rather than *merely* causal relationship to the mechanism as a whole.

The MM account promised to bridge the gap between philosophical analysis and scientific practice by showing how concrete experimental interventions could establish constitutive relations between mechanistic parts and wholes - thereby establishing that there is an empirical foundation for claims about mechanistic constitution that distinguish them from etiological causal claims.

2.1 The incoherence challenge

Among the criticisms faced by the MM account, one of the most daunting was the challenge of conceptual incoherence. A strong formulation of this challenge can be

[2] This proposal aligned with broader calls to connect philosophical accounts of biological mechanisms to the methodological practices of life scientists (e.g., Darden, 2006; Bechtel & Richardson, 1993/2010).

[3] While in Sects. 2 and 3 I rely on the mechanistic labels for 'interlevel' experiments (and the sub-categories of 'top-down' and 'bottom-up' experiments) for expository purposes, Sect. 5 will critically examine this widespread use.

found in Baumgartner and Gebharter (2016) and Baumgartner and Casini (2017), who argued that MM misapplies Woodward's interventionist analysis of causal relevance to a domain that precludes its proper adoption, rendering the account conceptually untenable.

The challenge centres on the requirements imposed by Woodward's framework for something to count as an *ideal* intervention (Woodward, 2003). For an intervention on variable X with respect to variable Y to count as ideal, the intervention must not change Y via any route other than X. Crucially, the intervention cannot change Y directly—if it does, then observed changes in Y might not be due to changes in X, but rather to other causal pathways. This constraint is viewed as essential to establishing genuine causal relationships rather than mere correlations.

The problem for the MM account is that this key condition cannot be satisfied when X and Y are also supposed to stand in part-whole relationships. If components are proper parts of the mechanisms they constitute, then one cannot intervene on the whole mechanism without simultaneously affecting its parts, and vice versa. In brief, the relationship between parts and wholes appears to make ideal interventions impossible.

Baumgartner and Gebharter (2016) make this problem sharper by identifying only three possible ways to understand interventions in interlevel experiments:

(I1) interventions on the whole which then change some specific part (problematic because such interventions violate the widely accepted prohibition on interlevel causation),
(I2) interventions on some specific part which then change the whole (problematic for the same reason - they violate the prohibition on interlevel causation), or.
(I3) interventions that are common causes of both the targeted part and the whole (problematic because they would violate the constraint that interventions not simultaneously manipulate putative causes and effects).

Since all three interpretations offered by Baumgartner and Gebharter violate core constraints of the interventionist framework, the MM account appears to have no viable way to use the Woodwardian framework to make sense of interlevel experiments as conclusively establishing constitutive relevance.

Romero (2015) developed a parallel charge of incoherence by framing the problem as a logical puzzle rather than focusing on specific intervention types. He argued that MM requires jointly satisfying four conditions that are logically incompatible:

1. parts and wholes are not causally related (no interlevel causation);
2. parts are proper components of wholes (part-whole relation);
3. ideal interventions can establish causal relevance (causal sufficiency); and.
4. MM requires interventions on wholes with respect to parts (interlevel intervention).

Romero's formulation suggests that the problem runs deeper than finding the right way to interpret interlevel experiments. According to his argument, the very attempt to apply interventionist methods to part-whole relationships leads to logical contradiction. Since conditions (1)-(4) cannot all be satisfied simultaneously, MM cannot offer a conceptually coherent analysis of mechanistic constitution.

The severity of the incoherence challenge was hard to ignore. Collectively, these arguments suggest that MM faces a crisis: the very experimental practices that were supposed to vindicate the constitutive/causal relevance distinction appear to undermine it, once constructed in interventionist terms. Moreover, this crisis seemed to demand more than just technical adjustments, requiring perhaps a wholesale reconsideration of (i) what constitutive relevance amounts to and (ii) how it can be experimentally established.

2.2 The matched interlevel experiments account

Craver et al. (2021) responded to this challenge by revisiting both the conceptual foundations of MM and the analysis of the experiments that are supposed to provide evidence for constitutive relevance relations within mechanisms. Their response involves a series of interconnected conceptual moves that ultimately reframe mechanistic constitution itself as a form of causal mediation, beginning with a fundamental change in how mechanisms are diagrammatically represented.

The first move in Craver et al.'s response involves replacing the traditional hierarchical representation of mechanisms (Fig. 1) with what they call a "flat" processual diagrammatic representation (Fig. 2). They explain this diagrammatic shift by noting that "ψ-ing is represented as a process beginning with an input, ψin, and terminating with an output, ψout. Between these temporal endpoints, and a mechanistic level down, is a temporally sequenced causal chain of events, involving the Xi and their various φi" (Craver et al., 2021, p. 8812).

This representational change, I argue, is not merely cosmetic–it reframes how mechanisms are to be understood. Rather than hierarchically organised part-whole

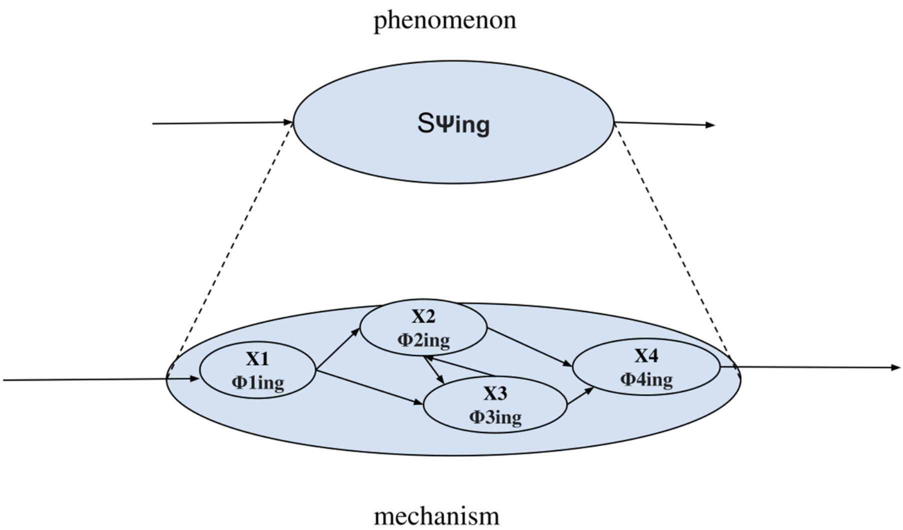

Fig. 1 A phenomenon and its mechanism (redrawn from Craver, 2009)

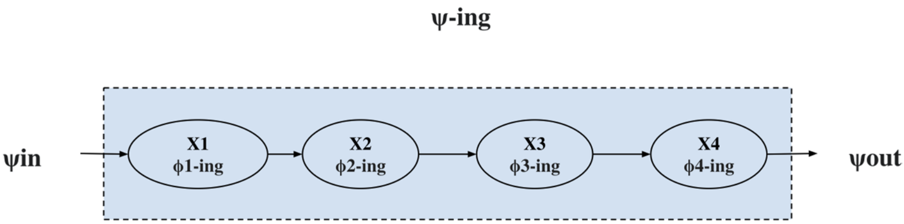

Fig. 2 The processual core of a mechanism (redrawn from Craver et al., 2021)

structures depicted as distinct levels, mechanisms are now represented as temporal processes. So the mechanistic diagram also shifts the main explanatory question from 'what parts constitute this mechanism?' to 'what causal steps bridge ψin and ψout?' The move from hierarchy to temporal sequence leads to a deeper conceptual transformation: the problem of establishing constitutive relevance becomes one of identifying causal pathways rather than finding part-whole relationships within mechanistic levels.

However, Craver et al. maintain that this representational change preserves mechanistic levels. They insist that the diagram's "flatness" is merely a representational artefact that maintains the distinction between (i) ψin and ψout and their relationship, which belong to a higher (phenomenal) mechanistic level, and (ii) the Xi's φj-ing and their causal relationships, which belong to a lower mechanistic level. This creates a tension: if the fundamental problem becomes identifying causal pathways (or what mediates between ψin and ψout) rather than part-whole relationships, what distinctive theoretical work can the notion of levels continue to perform?

Building on this processual reframing, Craver et al. (2021) make a decisive analytical move by reconceptualising interlevel experiments themselves. Rather than defending the criticised notion of interlevel interventions, they propose to interpret the same class of experiments as essentially three-variable causal mediation interventions. This move is enabled by the processual representation: once mechanisms are viewed as temporal sequences from ψin to ψout, experiments can be more straightforwardly understood as testing causal positioning within this sequence rather than as interventions that bridge mechanistic levels.

This transformation is captured in the new matched interlevel experiments (MIE) account:

> "(MIE) To establish that an entity X and its activity φ are constitutively relevant to a mechanism that ψs, the following experimental results and matching condition are jointly sufficient:
> (CR1i) If an experiment initiates conditions ψin while a bottom-up intervention, I, prevents or inhibits X's φ-ing, alterations to or prevention of ψ's terminal conditions, ψout, are detected.
> (CR1e) If a bottom-up intervention, I, stimulates X's φ-ing, ψ's terminal conditions, ψout are detected.
> (CR2)* If a top-down experiment initiates conditions ψin and detects ψ's terminal conditions, ψout, X's φ-ing is also detected.

(Matching) The activities φ activated or inhibited in bottom-up experiments (CR1i and CR1e) must be of the same kind and occur within quantitatively overlapping ranges with the activities φ detected in top-down experiments (CR2*).” (Craver et al., 2021: 8822).

Examining the structure of these conditions reveals how the original part-whole focus of the MM account has been transformed. Rather than testing whether components belong to mechanisms, (CR1i) and (CR1e) now test whether X’s φ-ing is causally necessary and sufficient for the ψin → ψout transition, while (CR2*) establishes whether X’s φ-ing is activated during this transition. The matching condition ensures that these constitute unified causal pathways rather than separate relationships.

Craver et al. make this conceptual transformation explicit when they conclude that these experiments test for ‘causal betweenness’, i.e., they show ‘that X and its φ-ing are caused by ψin, are causes of ψout, and are activated or changed in the process of producing ψout from ψin’ (Craver et al., 2021, p. 8823). Through this reformulation, experiments that were supposed to provide collective evidence for distinctive constitutive relevance relations have become experimental tests of causal mediation.

This reinterpretation allows Craver et al. to respond directly to the incoherence challenge. They claim that interlevel experiments should be understood as following the causal structure ‘Iψ → ψin → … → X’s φ-ing →… → ψout’ rather than as interventions that traverse any *purported* mechanistic levels. This represents a fourth option beyond the three problematic interpretations identified by Baumgartner and Gebharter (2016) which should dissolve the incoherence problem by treating interlevel experiments as non-problematic causal mediation experiments within an interventionist framework.

However, I claim that this solution comes at a significant theoretical cost. The MIE account explicitly departs from the original MM proposal when it claims that: ‘constitutive relevance amounts to a kind of causal mediation [since] ψin exerts its effect on ψout partly or wholly via an antecedent effect on X and its φ-ing’ (Craver et al., 2021, p. 8821). While the MM account was designed to show how experiments establish that specific part-entities stand in constitutive rather than merely causal relationships to whole mechanisms, the MIE account reconceptualises constitutive relevance as a form of causal relevance. This transformation raises a puzzle that I will pursue in the following sections: if constitution is causal mediation, what distinctive theoretical work can the concept of constitution continue to perform in mechanistic explanation?

3 Experimental evidence and the case for mechanistic constitution

This section evaluates the MIE account in terms of how well it withstands scrutiny when confronted with actual experimental practices in neuroscience. If Craver et al. are correct that interlevel experiments provide evidence for constitutive relevance through probing causal mediation relations, then their account should accurately describe how scientists conduct and interpret these experiments. However, a fair ‘test’ of Craver et al.’s proposal requires examining experiments that meet its own evidential requirements. According to the MIE account, constitutive relevance can only be established when bottom-up experiments, top-down experiments, and the matching condition are jointly satisfied—e.g., when the molecular activities manipulated in bottom-up interventions are the same as those detected in top-down experiments. To probe the account's descriptive adequacy, I will analyse experiments from action potential research that satisfy this matching requirement.

One clarification about my strategy in this section is in order. I am not arguing that these experiments somehow fail to establish what Craver et al. claim they establish. If Craver et al. are correct that (1) these experiments satisfy MIE's requirements, (2) MIE provides sufficient conditions for constitutive relevance, and (3) constitutive relevance is causal betweenness, then these experiments should reveal causal mediation relationships. My analysis demonstrates that this is precisely what they reveal - and crucially, that nothing beyond causal mediation is established. However, I take the "perfect fit" between these experiments and MIE's formal requirements to expose the theoretical reduction I flagged above: practices satisfying all of MIE's conditions establish sophisticated causal mediation without requiring any additional constitutive interpretation.

I focus on this area of experimental research for two main reasons. First, action potential experiments have already been discussed extensively in the mechanistic literature, with philosophers citing them as paradigmatic sources of evidence for mechanistic explanations (Craver, 2009; Levy, 2013; Kaiser & Krickel, 2017; Aizawa & Headley, 2022, 2025; Bickle, 2023). Second, contemporary action potential research employs well-developed experimental techniques that intuitively span what mechanists consider different "levels"—from molecular interventions on ion channels to cellular-level recordings of membrane potential changes. If the MIE account accurately describes any area of neuroscientific practice, it should be this one.[4]

My analysis begins with Hodgkin and Huxley (1952a, b, c, d; Hodgkin et al., 1952) seminal voltage clamp experiments, which I will take to exemplify top-down mechanistic interventions in action potential research. Their systematic investigation of ionic currents underlying neural excitability established the empirical foundation for understanding how molecular-level ion channel activity generates cellular-level electrical phenomena (Haüser, 2000; Catterall, Goldlin et al., 2012; Catterall, Raman, et al., 2012; Bickle, 2023)[5]. I then examine bottom-up experiments that manipulate

[4] Additionally, understanding action potentials mechanistically has been an important turning point in neuroscience (Clarke & Jacyna, 1987; Finger, 2000; McComas, 2011): these rapid electrical signals are seen as the fundamental unit of neural information processing, enabling everything from sensory perception to motor control to complex cognition. The processes underlying action potential generation, propagation, and modulation are therefore key to explaining how nervous systems function.

[5] In the full series of papers (Hodgkin & Huxley, 1952a, b, c, d; Hodgkin et al., 1952), the main research target is to identify the factors that control the movement of ions across excitable membranes and thus give rise to the action potential. Identifying the effects of these factors on ionic flux and describing them quantitatively was seen as facilitating not only conceptual understanding of electrical signaling but also reliable predictions of how an excitable membrane would behave under any set of novel conditions. To achieve this end, the scientists had to determine first whether sodium and potassium were indeed the ions whose flux generated the voltage change that constituted the action potential (Raman & Ferster, 2022; Bickle, 2023). One could argue that it is this focus on the "identity" of the ions involved in the mechanisms of the action potential which further supports the choice of this case study as very well suited for understanding the experimental methodologies that might support constitutive claims.

the same potassium channel activities that Hodgkin and Huxley detected in their experimental work, establishing the matching that MIE requires for establishing constitutive relevance. Do these experiments establish constitutive relationships between molecular components and cellular processes, or do they only support causal mediation hypotheses?

3.1 Hodgkin and huxley's voltage clamp experiments as Top-down interventions

In this section, I analyse Hodgkin and Huxley's (1952a, b, c, d; Hodgkin et al., 1952) voltage clamp experiments on the squid giant axon as a paradigmatic example of a top-down mechanistic intervention. Their experimental approach involved controlling membrane voltage while measuring the ionic currents flowing across the cell membrane. This intervention broke down what now is known to be the normal feedback cycle between voltage changes and ionic currents that characterises action potential generation. Using an innovative two-electrode variant of the voltage clamp—with separate recording and stimulating electrodes inserted into the squid giant axon at different locations[6]—Hodgkin and Huxley relied on electronic feedback to hold membrane voltage constant at predetermined levels. This enabled them to isolate and then systematically investigate the ionic currents that would normally drive voltage changes in action potentials generated in normal conditions.

The structure of their experimental protocol can be taken to exemplify top-down interventions for several reasons: Hodgkin and Huxley manipulated a cellular-level phenomenon (membrane voltage) while detecting molecular-level activity (ionic currents). Summarised in their five landmark papers (Hodgkin & Huxley, 1952a, b, c, d), their experimental work involved the painstaking refinement of experimental controls. Specifically, Hodgkin and Huxley systematically varied membrane voltage from resting potential (−60mV) to various depolarised levels (+ 50mV and beyond), creating controlled conditions under which they could measure how voltage changes trigger specific ionic currents.

These controlled conditions were specifically designed to test the *sodium hypothesis*—the idea that action potentials result from a large and selective increase in membrane permeability to sodium ions (Na+), followed by slower efflux of potassium ions (K+). Linking the experimental interventions to this hypothesis allows me to delineate their three-variable structure: membrane voltage steps (the independent variable) caused systematic changes in ion channel activation states (the intermediate variable), which in turn produced measurable ionic currents (the dependent variables).

Through a series of ionic substitution experiments—systematically altering Na + ion concentrations in the nutrient bath—Hodgkin and Huxley were able to characterise the distinct kinetics and voltage-dependence of sodium and potassium currents separately. More specifically, their experimental manipulations provided evi-

[6] The technical advances that allowed Hodgkin and Huxley to control the membrane potential of a short stretch of the axon (i.e., to voltage-clamp the membrane) were first presented in Hodgkin et al. (1952) paper, which also reported the first recordings of ionic currents at a series of experimentally determined, fixed membrane potentials.

dence that sodium currents were “fast but transient” (roughly 1.5 milliseconds) while potassium currents were “delayed but longer-duration” (less than 2.5 milliseconds). These differences in temporal dynamics supported the hypothesis of independent molecular mechanisms underlying the distinct sodium and potassium conductances (Fig. 3).

This experimental structure—controlling cellular variables to detect molecular responses—fits MIE’s definition of top-down experiments. Yet the methodology consistently treats voltage and current as causally related variables rather than as part and whole. Crucially, Hodgkin and Huxley framed their hypotheses in explicitly causal terms throughout: they investigated how membrane depolarisation causally drives sodium channel activation, how sustained depolarisation causally drives potassium channel activation, and how these ionic currents causally contribute to the membrane voltage changes observed in natural action potentials. Their experimental methodology consistently treated voltage changes as causes and ionic currents as effects, establishing causal dependencies rather than testing hypotheses about part-whole relationships[7].

These top-down experiments establish one half of MIE’s evidential requirements. To complete the experimental matching that MIE demands, we need to look for bottom-up experiments that manipulate the same molecular activities Hodgkin and Huxley detected. This would provide a more decisive evaluation of whether mechanistic interventions can reliably distinguish constitutive relationships from causal dependencies.

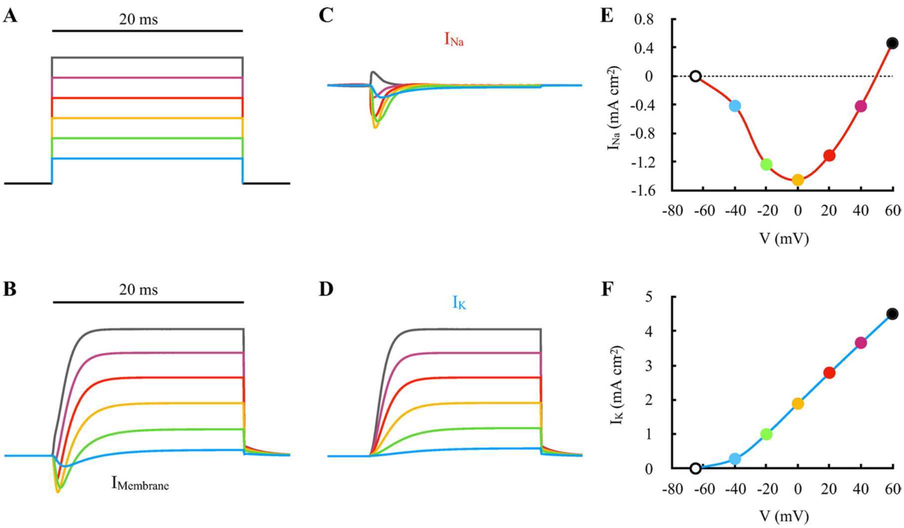

Fig. 3 This data is mandatory please provide

[7] Bickle (2023) has recently pointed out that Hodgkin and Huxley explicitly characterised their celebrated quantitative model as “nothing more than an empirical description” and acknowledged that “different mechanisms could lead to similar equations.” The recognition that multiple distinct mechanisms could satisfy the same causal relationships further strengthens my argument that their work focused on investigating causal rather than constitutive relationships.

3.2 Bottom-up experiments

The bottom-up experiments I examine in this section target precisely the potassium channel activities that Hodgkin and Huxley characterised[8], providing the experimental symmetry MIE requires for constitutive relevance claims. These experiments involve manipulating (inhibiting or exciting) molecular factors and detecting effects at the cellular level. Because of the difference in spatial scale between the entities whose properties are experimentally manipulated (molecules) and those whose properties are detected and measured (cells), they intuitively fit into Craver et al.'s category of bottom-up experiments.

According to MIE, these experiments should establish constitutive relevance because they demonstrate that molecular components are 'proper' parts of cellular mechanisms through coordinated experimental protocols. If MIE were successfully distinguishing constitutive from merely causal relationships, we would expect these experiments to reveal something beyond sophisticated causal mediation—perhaps evidence of part-whole organisational features, synchronous dependence relationships, or compositional dependencies that differ from temporal causal sequences. However, as I will show, the experimental logic consistently follows causal mediation patterns.

As the first step in this analysis, I examine Frankenhaeuser and Hodgkin's (1956) experiments using elevated external potassium concentrations to enhance delayed rectifier K + channel activity. This approach targeted the same channels that Hodgkin and Huxley had previously characterised in their voltage clamp work. The experimental protocol of the 1956 study also appears to follow the mechanistic bottom-up structure proposed by Craver et al. (2021): the researchers manipulated molecular-level factors (ionic gradients affecting K + channel activity) while detecting cellular-level responses (action potential characteristics).

More specifically, Frankenhaeuser and Hodgkin systematically varied the extracellular K + concentration in the bathing solution surrounding squid giant axons, creating enhanced electrochemical driving forces for K + efflux through the delayed rectifier channels. The experimental procedure involved preparing nerve tissue samples, establishing baseline recordings in normal saline, then switching to solutions with progressively higher [K+] concentrations while monitoring action potential changes. When they triggered action potentials under these enhanced ionic conditions, they observed systematic and predictable alterations in electrical activity.

The experimental results demonstrated that elevated external [K+] produces distinctive changes in action potential characteristics: faster repolarisation, shortened action potential duration, and more pronounced afterhyperpolarisation phases. These changes resulted from amplified delayed rectifier K + currents—the same molecular activity that Hodgkin and Huxley had detected in their voltage clamp experiments.

[8] Potassium channel activity was not the only variable investigated in Hodgkin and Huxley in their experimental work, but I focus on it as a selection criterion for identifying "matching" bottom-up interventions. For a comprehensive discussion of the experimental work on which Hodgkin and Huxley built their quantitative model of voltage-dependent conductances, see Raman and Ferster (2022).

Analysing these excitatory experiments from 1956 reveals a clear causal mediation structure: elevated external [K+] (independent variable) enhances the electrochemical driving force for channel activity (intermediate variable), producing measurable changes in action potential kinetics (dependent variables). The dose-response relationships that Frankenhaeuser and Hodgkin established—showing that higher external [K+] produces correspondingly greater effects on action potential characteristics—demonstrate a causal pathway from molecular manipulation to cellular-level response. The ‘matching’ MIE requires is satisfied, but only at the level of causal variable identification.

To create the experimental symmetry that MIE requires, I next consider Armstrong and Hille’s (1972) experiments using tetraethylammonium (TEA) to block the same K+channels. TEA acts as a specific blocker by binding to the intracellular opening of these potassium channels, effectively preventing the outward flow of K+ions that Hodgkin and Huxley had shown was responsible for action potential repolarisation.

Armstrong and Hille’s experimental protocol involved several concrete steps that demonstrate the bottom-up structure of their intervention. First, the researchers prepared nerve tissue samples and established baseline recordings of normal action potential waveforms using microelectrodes. Next, they applied TEA to the bathing solution surrounding the nerve tissue, allowing the drug to penetrate and bind to the potassium channels. Finally, they triggered action potentials using electrical stimulation and recorded the resulting changes in electrical activity.

When TEA blocked the K+channels, it produced systematic alterations directly opposite to the enhancement effects: the repolarisation phase became significantly prolonged, the action potential duration increased dramatically, and the membrane took much longer to return to its resting voltage. This experimental approach complements both Hodgkin and Huxley’s top-down strategy and Frankenhaeuser and Hodgkin’s excitatory bottom-up approach. While the former controlled cellular-level voltage to detect molecular-level ionic currents, and the latter enhanced the driving forces for those currents, Armstrong and Hille eliminated the currents entirely through selective channel blockade.

Analysing the structure of their experimental procedure reveals the same three-variable causal pattern: TEA application (independent variable) causes selective blockade of delayed rectifier channels (intermediate variable), which produces measurable changes in action potential characteristics such as prolonged duration and slowed repolarisation kinetics (dependent variables). The dose-response relationships demonstrated that higher TEA concentrations produce more complete channel blockade and correspondingly greater action potential prolongation, supporting the existence of a causal pathway from molecular intervention to cellular-level response.

Both experimental studies framed their research questions in explicitly causal terms. Frankenhaeuser and Hodgkin investigated how enhanced K+gradients causally amplify repolarisation currents, while Armstrong and Hille examined how channel blockade causally alters action potential repolarisation and how different drug concentrations produce dose-dependent effects on electrical excitability. Despite manipulating molecular components and detecting cellular-level responses—precisely the spatial scale relationship that MIE suggests should constitute evidence for

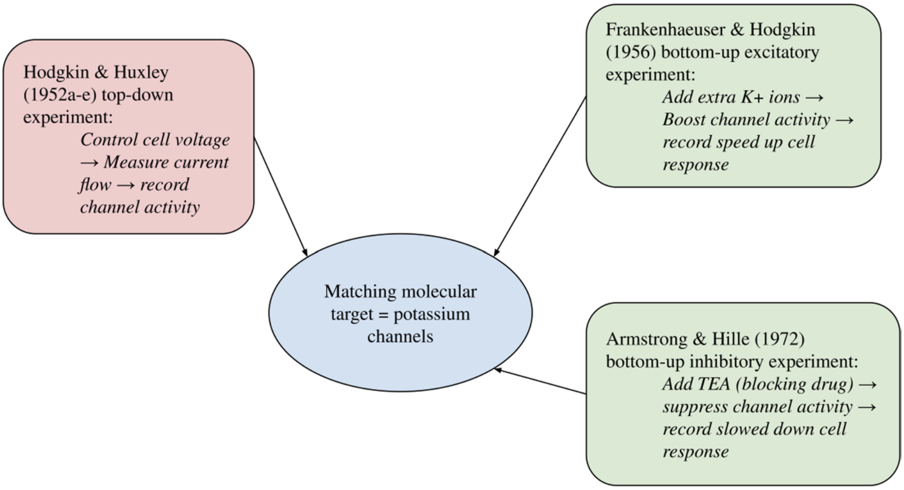

Fig. 4 A diagrammatic representation of the "matched" experiments analysed in section 3

part-whole relationships—both methodologies remain focused on establishing causal dependencies rather than testing constitutive relationships.

The previous analysis reveals that all three experimental approaches, selected because they satisfy MIE's matching requirements (Fig. 4), consistently provide evidence for causal mediation relations. Now, if Craver et al. (2021) are correct and there is nothing more to constitution than causal betweenness, then we wouldn't expect these experiments to provide any additional information about the structure of action potential mechanisms. However, is this conceptual transformation realised by the MIE account a satisfactory analysis of mechanistic constitution? How do these experiments support genuine constitutive claims?

4 On the theoretical status of constitutive categories

4.1 Lessons from action potential experiments

The analysis of action potential research in Sect. 3 showed that experiments satisfying MIE's matching requirements identify systematic three-variable causal mediation relationships. I have argued that whether controlling voltage to detect ionic currents (Hodgkin & Huxley, 1952a, b, c, d; Hodgkin et al., 1952), enhancing channel activity to amplify cellular responses (Frankenhaeuser & Hodgkin, 1956), or blocking channels to eliminate responses (Armstrong & Hille, 1972), the experimental protocols consistently aim to test whether molecular factors causally mediate between measurable cellular inputs and outputs.

These experiments provide compelling evidence for causal mediation relationships while satisfying MIE's formal conditions. Potassium channel activity is necessary for normal action potential repolarisation (CR1i), sufficient to drive repolarisation when enhanced (CR1e), and consistently activated during normal action potential generation (CR2*). The matching condition is satisfied because the same delayed rectifier potassium channel activity appears across all experimental approaches. Yet this 'perfect' fit also reveals that the experiments which establish causal mediation don't require any additional theoretical interpretation in constitutive terms.

Thus I claim that the previous analysis exposes a core tension in Craver et al.'s philosophical project. If some of the most sophisticated experimental protocols designed to detect constitutive relationships reveal only causal mediation, either (1) the experimental approach is inadequate for detecting genuine constitution, or (2) constitutive categories lack the distinctive theoretical content mechanistic philosophers claim.

4.2 Two types of constitution?

Before exploring whether stronger theoretical arguments can recover the distinctiveness of constitutive categories, I should address a potential response to the analysis in Sect. 2. For one might argue that the apparent reduction of constitution to causal mediation reflects not a theoretical problem but rather a clarification strategy. That is, one might claim that there are two different types of constitution at play (entity and process constitution), and the MM→MIE transformation marks the movement from a conflation of the two categories to the proper focus on the category of process constitution.

On this interpretation, entity constitution involves spatial parts in synchronous part-whole relationships (the heart is part of the circulatory system; the ALML neuron is part of the worm's nervous system), whereas process constitution involves temporal stages in sequential relationships (the rising phase is part of the action potential; repolarisation is part of neural signalling). Craver (2007a, b, c) conflated these in MM; critics like Baumgartner and Gebharter (2016) exposed the confusion; Craver et al. (2021) resolved it by focusing MIE on process constitution while dropping the "X is part of S" requirement concerning entity constitution (Craver et al., 2021, footnote 4).

However, I argue that such a "two types of constitution" interpretation can strengthen rather than undermine my main thesis. First, if we accept this interpretation, the MIE account seems to have abandoned the notion of entity constitution—the spatial-compositional relationships that motivated the original mechanistic program. Machamer et al. (2000), and Craver (2007a, b, c) emphasised hierarchical organisation and how spatial arrangements of components link to system-level capacities. The "X is part of S" requirement attempted to capture this. By dropping it, MIE changes what mechanistic explanations are meant to accomplish: they now map temporal sequences rather than compositional structures.

Second, and more critically, even granting process constitution as a legitimate category, one is left wondering if it provides any distinctive theoretical content beyond causal mediation once we understand it through Craver et al.'s framework. The question is not whether processes have temporal parts—they do—but what theoretical work "constitution" performs once Craver et al. explicitly identify it with causal mediation.

Consider their explicit formulations: constitutive relevance "amounts to a kind of causal mediation" (2021, p. 8821); the problem is "identifying the components of the process bridging ψin and ψout: What lies on the causal path(s) between these phenomenon-defining endpoints?" (2021, p. 8812); MIE experiments "add up to sufficient evidence that X and its φ-ing lie causally between ψin and ψout" (2021, p. 8823). The metaphysical basis for constitutive relevance is also explicitly identified with causal betweenness. So the driving question of the present argument still stands. What distinguishes "X's φ-ing is constitutively relevant to ψ-ing" from "X's φ-ing causally mediates between ψin and ψout"? The former might signal pragmatically that we're discussing mechanistic components rather than external triggers. But I contend that utility (suggesting a convenient label for a causal subset) might not yet establish theoretical distinctiveness.

The "two-types-of-constitution" interpretation thus presents a dilemma: either (1) process constitution differs theoretically from causal mediation, making Craver et al.'s identification of constitutive relevance with causal betweenness problematic; or (2) process constitution is adequately captured as causal mediation, making "constitution" terminological shorthand for relationships mediating between phenomenally-defined endpoints rather than a fundamental theoretical category.

Craver et al.'s theoretical commitments implicitly point to option (2). The consistency of their solution to the incoherence challenge depends on this reduction. My analysis makes explicit what their framework implies: once mechanisms are understood processually and constitutive relevance defined as causal betweenness, constitutive categories become theoretically redundant.

4.3 The strengthened case for mechanistic constitution

Nevertheless, even accepting this reduction, one might argue constitutive categories retain distinctive *explanatory* value. Perhaps MIE's coordinated experimental methodology—requiring inhibitory, excitatory, and detection experiments plus matching conditions—reveals organisational features that form an explanatorily significant category. Or perhaps mechanistic explanation's traditional targets justify constitutive in addition to a purely causal treatment.

Fagan (2015) developed two lines of argument for defending the explanatory value of constitutive relevance. First, she argued that there is a key explanandum distinction that separates causal and constitutive mechanistic explanations. Whereas causal explanations answer etiological questions (What causes phenomenon P?) where the explanandum is an effect produced by a specific mechanism, constitutive mechanistic explanations answer operational questions (How does mechanism M work?) where the explanandum is the working mechanistic system itself. She claimed that the latter are pertinent in investigating "many biological processes […]: growth, reproduction, infection, blood circulation, development, metabolism, aging, death… In these inquiries, the question to be answered is: 'how does mechanism M work?'" (Fagan, 2015: 72). This explanandum distinction suggests that constitutive and causal explanations target genuinely different aspects of biological phenomena—system organisation rather than causal production—not merely different descriptions of the same causal relationships.

Second, Fagan provided a complementary organisational argument. She argued that mechanistic components have "meshing properties" that determine specific organisational patterns irreducible to individual causal contributions: "[constitutive] explanations do not describe each component's individual contribution to the system behaviour. Rather, the explanatory model displays how all the interrelated parts together constitute the system behaviour" (Fagan, 2015: 74). Fagan developed this through her notion of "jointness"—components jointly constitute system behaviour when they form a complex in virtue of meshing properties such that when united as a complex the components perform a specific detectable activity, but not otherwise. This suggests that constitution involves symmetric binding relations and organisational features that cannot be captured through standard causal analysis focused on asymmetric cause-effect relationships.

Taken together, these arguments represent the strongest available case for recovering constitutive categories on explanatory grounds. The explanandum distinction indicates that constitutive explanations target system organisation rather than causal production. The organisational argument provides theoretical grounds for why this matters—mechanisms have complex organisational patterns that cannot be completely captured through analysis of individual causal pathways. If successful, these arguments would show that the experimental practices I analysed in Sect. 3 miss crucial organisational features determining how components work together as unified systems.

However, I contend that the MM→MIE transformation poses a serious challenge to these enhanced explanatory motivations. The reduction of constitutive relevance to causal mediation threatens to undermine both Fagan's explanandum distinction and organisational argument. This challenge requires careful formulation, particularly since Fagan's own examples support a processual understanding of mechanisms. Her paradigmatic cases—growth, reproduction, blood circulation, aging, death (2015, p. 72)—are easily conceivable as processes unfolding over time rather than static systems or spatial structures. This lends initial plausibility to Craver et al.'s (2021) processual reframing. If mechanistic explanation's central targets are inherently temporal phenomena, perhaps process constitution is exactly what mechanistic philosophy should analyse. The "how M works" questions that Fagan identifies as distinctively constitutive seem naturally suited to these processual targets.

The key issue is whether process constitution—once explicitly understood as causal betweenness—preserves what made the constitutive/causal distinction explanatorily valuable. I argue it does not. Consider first Fagan's explanandum distinction between "What causes P?" and "How does M work?" Craver et al.'s processual representation explicitly reconstructs the latter in causal terms: "the problem of constitutive relevance is that of identifying the components of the process bridging ψin and ψout: What lies on the causal path(s) between these phenomenon-defining endpoints?" (2021, p. 8812, emphasis added). The central problem becomes identifying causal pathways.

If constitutive relevance is causal betweenness, then answering "How does M work?" reduces to mapping causal pathways mediating between ψin and ψout. Understanding how M works becomes identifying what causal steps bridge ψin and ψout. Mechanisms are "temporally sequenced causal chain[s] of events" (2021, p.

8812); even feedback cycles “should be understood, sequentially in time” (2021, p. 8810).

The distinction between explaining organisation versus explaining production collapses when organisation is reconceptualised as temporal causal sequence. Under MIE, explaining “how M works” becomes explaining “what M causes” in finer temporal detail—both map causal dependencies, with constitutive explanations revealing additional mediation steps. Fagan’s explanandum distinction may reflect pragmatic differences (system-level behavior versus triggers), but both receive the same answer type: descriptions of causal pathways.

Fagan’s organisational argument fares no better. A careful reader might note that Craver et al. emphasise organisational complexity, describing mechanisms as “rope[s] of densely entwined fibers” where “the same, particular components can appear time and again at different stages” (2021, p. 8810). This seems to capture Fagan’s “meshing properties” and “jointness.” However Craver et al. also add: “such relations, properly understood, can be represented, and should be understood, sequentially in time” (2021, p. 8810). Organisational patterns get temporalised into causal sequences. The processual representation advocated on the MIE account treats organisational features as arrangements of causal interactions over time rather than distinctive constitutive dependencies.

Even Fagan’s “symmetric binding relations” and “meshing properties” are accommodated within MIE as features of causal sequences—dispositions for temporal interactions, dependencies enabling $\psi in \rightarrow \psi out$ transitions, constraints on causal pathways. These are important mechanistic features, but they can be characterised as properties of complex causal systems. On the MIE account, therefore, Fagan’s jointness reflects causal interdependencies within temporal processes rather than a categorically distinct compositional mode.

MIE implies that Fagan’s organisational irreducibility is actually complex causal structure capturable through sophisticated causal analysis—precisely what three-variable mediation experiments demonstrate. Section 3 showed how practices satisfying MIE’s requirements establish intricate causal mediation patterns. Potassium channel activity is necessary for repolarisation (causal necessity), sufficient when enhanced (causal sufficiency), and consistently activated during action potentials (causal intermediacy). The matching condition ensures unified causal pathways. This is sophisticated causal analysis of complex biological systems, but sophistication doesn’t create categories beyond causal mediation.

The MM→MIE transformation thus shows that even sophisticated defenses of mechanistic constitution face substantial pressure from causal mediation interpretations. Fagan’s case aimed to show that constitutive categories capture something irreducible to causal analysis—either through distinctive explananda or through appealing to organisational features. However, once constitutive relevance is explicitly defined as causal betweenness and mechanistic processes understood as temporal causal sequences, the theoretical work constitutive categories were supposed to perform gets absorbed into sophisticated causal analysis of complex, multi-scale temporal structures.

This analysis demonstrates a theoretical reduction: mechanistic constitutive categories lack distinctive content beyond what sophisticated causal mediation analysis captures. One final objection, however, merits examination. Perhaps mechanistic levels provide grounds for preserving constitutive categories even if individual constitutive relations can be analysed without remainder to causal mediation. According to this objection, the hierarchical, multi-level organisation of mechanisms reflects a distinctive organisational structure that cannot be captured in terms of temporal causal sequences alone.

5 Can mechanistic levels vindicate constitutive categories?

Defenders of constitutive categories might appeal to mechanistic levels in two ways. On a traditional approach, levels are to be understood in terms of compositional hierarchy and spatial scale: molecular ion channels compose cellular membranes, which compose neural circuits. Thus discrete organisational levels are connected by constitutive relations. More recent philosophical accounts of levels (Wimsatt, 2007; Potochnik & McGill, 2012; DiFrisco, 2017) avoid naive compositional assumptions but still seek to preserve the explanatory work that the notion of levels performs in accounts of how the world is organised. I examine both versions, showing why neither can provide the analytic resources that Craver et al. need for a more substantive notion of mechanistic constitution.

5.1 Hierarchical levels and experimental practice

The traditional formulation grounds levels in hierarchical compositional structure based on spatial scale and part-whole relationships (Oppenheim & Putnam, 1958; Simon, 1962). Craver et al. themselves seem to rely on this notion when they insist their processual representation preserves mechanistic levels: "While some might take Fig. 2 to do away with levels, it does not" (2021, p. 8811–8812). They argue that ψ-ing remains at a higher level than the various φi-ings even in the processual framework, with levels defined through constitutive relevance relations rather than spatial metaphors of "above and below."

In the case of the electrophysiological experiments analysed in Sect. 3, mechanistic levels would map onto spatial scale relationships between membrane voltage (cellular-level) and ionic currents (molecular-level), exemplifying the hierarchical organisation of action potential mechanisms. The objection contends that this hierarchical structure has organisational relationships that temporal causal mediation cannot fully capture—there remains a genuine distinction between the phenomenon as a whole (the action potential) and its mechanistic components (ion channel activities), even when both are understood processually. In response, I argue that appeals to hierarchical levels become problematic when examined against both the logic of Craver et al.'s framework and actual experimental practices.

Consider again the voltage clamp experiments that Hodgkin and Huxley (1952a, b, c, d; Hodgkin et al., 1952) used to investigate action potential mechanisms. The voltage clamp apparatus functions simultaneously as both an intervention device (controlling membrane voltage) and a measuring device (detecting the compensatory current required to maintain that voltage). Crucially, because this compensatory current must equal the magnitude of ionic currents generated by individual channel activities, the 'macro-level' voltage measurements necessarily contain information about 'micro-level' molecular processes. More importantly, both the controlled voltage and the detected currents probe what are essentially comparable entities: electrical properties of membrane patches, differing primarily in the spatial extent of measurement rather than representing genuinely distinct organisational levels (see also Baetu, 2025)[9].

Similarly, the bottom-up experiments involving TEA blockade and enhanced potassium gradients do not actually manipulate individual molecular components as the levels interpretation suggests. These interventions affect subpopulations of channel types through bulk manipulations—altering the average contribution of all channels sharing the same protein structure and conductance properties rather than targeting individual molecular entities.

The spatial scale relationships that appear to support hierarchical organisation actually reflect methodological differences in the scope of experimental manipulation rather than categorical organisational distinctions between mechanistic levels. When the mechanistic diagram shifts from Fig. 1's hierarchical representation to Fig. 2's processual representation, more changes than Craver et al. acknowledge. The shift transforms the explanatory question from "what parts compose this system?" to "what causally mediates between these temporal endpoints?" Once this transformation takes place, the notion of mechanistic levels loses its distinctive theoretical content.

Craver et al.'s own language reveals this tension. They acknowledge that "the spatial metaphor of above and below can sometimes make this relation sound much more mysterious than it in fact is" (2021, p. 8812). But the issue is not merely that spatial metaphors are misleading—it is that once we remove the spatial, synchronous, hierarchical understanding of levels, what remains is simply temporal positioning within causal sequences. The "higher level" ψ-ing is temporally bounded by ψin and ψout; the "lower level" φi-ings are causally intermediate between these endpoints. "Level" becomes a label for temporal position in a causal process rather than a marker of a different kind of hierarchical organisation.

5.2 Heuristic work for mechanistic levels?

At this point, one might object that even if hierarchical levels fail to match experimental practice, levels-talk still serves important heuristic functions in scientific fields like biology. Perhaps constitutive categories remain valuable because organising explanations around levels—cellular, molecular, atomic—provides useful conceptual scaffolding for scientific research. Scientists successfully employ levels vocabulary to organise findings, structure inquiries, and communicate complex phenomena. This persistence of 'levels-talk' in scientific practice might vindicate con-

[9] While in Sect. 3 I focused on showing that these "top-down" experiments establish causal mediation relationships, the levels objection asks one to reconsider whether the spatial scale differences involved might nevertheless support genuine hierarchical organisation and thus constitutive relations.

stitutive categories on pragmatic grounds even if levels don't map ontologically onto sharp compositional boundaries.

This objection shifts the grounds of debate. Rather than defending levels as objective features of nature, it foregrounds their explanatory utility. Perhaps "cellular level" and "molecular level" function as useful organisational concepts even if the boundaries are vague and the voltage clamp experiments show overlapping scales. The multi-level vocabulary might help scientists navigate complex causal structures without requiring that levels mark ontologically distinct organisational strata.

In response, I claim that this kind of heuristic utility does not suffice to establish theoretical distinctiveness for mechanistic constitutive categories. To unpack this claim, I will draw on a recent argument developed by Potochnik (2021) which shows that scientists' use of levels-talk reflects pragmatic organisational choices rather than objective features of nature. Her analysis will help me strengthen my point that mechanistic levels may function as useful heuristics while failing to ground objective distinctions between constitution and causation.

The core of Potochnik's argument is that different criteria for identifying levels—composition, scale, causal autonomy, domains of different sciences—don't align with each other. What counts as a "level" on compositional grounds differs from what counts as a level on causal or disciplinary grounds. Moreover, even within a single criterion, boundaries often remain vague and context-dependent. Applied to the action potential case, one might identify levels compositionally (atoms, molecules, channels, membranes), by spatial scale (nanometers, micrometers), or by causal dynamics (bonding, protein folding, voltage dynamics). These decompositions cross-cut in complex ways but they don't point to one single privileged hierarchical organisation.

The voltage clamp experiments exemplify this. The membrane voltage is simultaneously a cellular-level phenomenon (distributed across the membrane), a molecular-level phenomenon (arising from ionic distributions), and a biophysical phenomenon (electrical field across lipid bilayers). There's no fact of the matter about which level it belongs to—it participates in multiple overlapping patterns. This is precisely what makes levels useful heuristically: scientists can flexibly organise the same phenomena in different ways depending on explanatory interests.

However, this flexibility undermines rather than supports using the notion of level to ground objective distinctions between constitution and causation. If levels are perspectival organisational tools—ways of carving up causal structure to suit particular explanatory purposes—then they cannot provide objective grounds for claiming that some relationships are constitutive while others are merely causal. The choice to describe ion channels and action potentials as being at "different levels" reflects pragmatic convenience rather than discovering categorically distinct modes of dependence.

Potochnik's positive proposal reinforces this point. She argues that instead of reifying levels, we should recognise that scientists map "causal patterns"—regularities in how things influence each other. These patterns "can be organized and investigated in multiple ways depending on explanatory interests, without privileging any particular decomposition as the 'true' levels" (p. 78). This aligns with my analysis throughout this paper: mechanistic explanations reveal complex causal structures that can be organised in multiple ways. "Constitution" marks pragmatic choices about how to organise this causal information rather than tracking relationships fundamentally different from causation.

This interpretation also helps me account for the tension in Craver et al.'s framework identified in Sect. 5.1. They want to preserve levels-talk while adopting a processual, temporalised representation of mechanisms. But levels presuppose relatively stable, discrete organisational boundaries—exactly what processual representations undermine. Processes flow continuously; levels require discrete boundaries. The awkwardness of insisting levels remain while acknowledging that spatial metaphors are misleading reflects the deeper problem that their processual framework has made levels-talk untenable. If Potochnik is right that levels-talk is perspectival, then we can see why the processual reframing creates such difficulties: Craver et al. are trying to preserve heuristic concepts as if they marked objective organisational features.

Scientists organise information using many useful fictions—frictionless planes, ideal gases, average organisms, genetic "programs." These serve important heuristic and pedagogical functions without picking out genuine ontological categories. If levels are similarly perspectival tools, then constitutive language tied to levels becomes a useful convention rather than tracking categorically distinct relationships. The burden remains on defenders of constitutive categories to show why levels-talk should be treated differently—why it marks genuine theoretical distinctions rather than pragmatic choices.

In summary, appeals to the persistence of levels-talk cannot vindicate constitutive categories on theoretical grounds. Section 5.1 showed that traditional hierarchical levels fail to match experimental practice—the voltage clamp experiments don't achieve genuine scale separation, and apparent level-differences reflect methodological scope rather than categorical organisational distinctions. Even granting that levels serve useful heuristic functions, perspectival utility doesn't establish theoretical distinctiveness. If levels are flexible tools for organising causal information rather than objective features of nature, they cannot ground theoretical distinctions between constitution and causation. So, neither traditional compositional hierarchy nor appeals to explanatory utility are able to provide the analytic resources Craver et al. need for a substantive notion of mechanistic constitution.

6 Conclusions

I have argued that the transformation from the mutual manipulability (MM) to the matched interlevel experiments (MIE) account successfully resolves the incoherence challenge that plagued MM, but it does so at the cost of eliminating the distinctive theoretical content that constitutive categories were supposed to provide in mechanistic explanations.

My analysis wove together four complementary lines of argument. First, I showed that the conceptual transformation from MM to MIE reveals how attempts to preserve the coherence of mechanistic constitution within an interventionist framework eliminate its distinctive features. The shift from hierarchical to processual representations, the reinterpretation of interlevel experiments as three-variable causal mediation tests, and the explicit reconceptualisation of constitutive relevance as causal betweenness collectively demonstrate a theoretical reduction. Mechanistic explanations, according to MIE, map sophisticated causal mediation patterns rather than revealing categorically distinct constitutive structures.

Second, this conceptual analysis gains support from examining paradigmatic neuroscientific experiments—from Hodgkin and Huxley's voltage clamp studies to contemporary channel manipulation experiments. I showed that practices satisfying MIE's formal requirements consistently establish causal mediation relationships. Despite targeting the same molecular activities across different experimental approaches and satisfying the matching conditions that MIE requires for constitutive relevance, these studies reveal systematic three-variable causal structures without requiring constitutive interpretation.

Third, I addressed the interpretive objection that this reduction merely clarifies two types of constitution (entity versus process). Even granting this distinction, process constitution as defined by Craver et al. provides no distinctive theoretical content beyond causal mediation once explicitly identified with causal betweenness.

Fourth, I showed that even sophisticated defences of mechanistic constitution—including Fagan's (2015) explananda distinction and organisational arguments—face substantial pressure from causal mediation interpretations. The processual reframing proposed by MIE undermines both the distinction between explaining system organisation versus causal production and arguments about organisational irreducibility.

Finally, I addressed a structural objection arguing that appeals to mechanistic levels cannot rescue constitutive categories. Neither traditional hierarchical levels (which fail to match experimental practice) nor the argument for the heuristic utility of levels-talk (which reflects perspectival organizational choices rather than objective features) can ground theoretical distinctions between constitution and causation.

In questioning the theoretical necessity of constitutive categories, this analysis does not challenge the continued use of mechanistic explanatory frameworks in neuroscience. Instead, it suggests that mechanistic philosophy might benefit from developing a more precise conception of mechanisms as a theoretically distinctive subclass of causal explanations, clarified through attention to the specific types of causal structure and experimental methodology they involve.

Rather than undermining mechanistic approaches, this theoretical reduction points toward more precise and theoretically grounded directions for mechanistic philosophy. If constitutive categories are theoretically redundant, how should mechanistic philosophy characterise its distinctive contribution? Two complementary research programs emerge from this analysis, both of which preserve explanatory insights while eliminating theoretical confusion.

One promising direction involves developing a more precise characterisation of the types of causal systems that mechanistic explanations target. Rather than distinguishing mechanistic from causal explanations, we might distinguish mechanisms from other causal concepts like pathways, cascades, circuits, and networks based on their specific organisational features, temporal dynamics, or methodological requirements (Ross, 2021). This approach aligns with Ross and Bassett's (2024) claim that neuroscience employs "a rich causal language" where different terms capture distinct types of causal systems, suggesting that mechanistic philosophy should engage more systematically with this conceptual diversity rather than defending an overly broad mechanistic umbrella.

A second complementary direction is suggested by the processual reframing developed in the MIE account, which indicates that mechanistic explanations might be better understood through temporal causal structures rather than hierarchical organisational relationships. This raises questions about whether traditional causal concepts adequately capture the complex, multi-scale, and temporally extended causal patterns that characterise biological systems. Developing more sophisticated causal categories that accommodate features like synchronous dependence, organisational constraints, and multi-level integration might preserve the explanatory insights of mechanistic approaches without requiring non-causal constitutive relationships.

In summary, I suggest that mechanistic approaches in neuroscience derive their explanatory power not from distinctive constitutive relationships, but from their methodological sophistication in investigating complex, multi-scale causal structures. Theoretical clarity about what mechanistic explanations actually accomplish—revealing sophisticated causal organisation rather than categorically distinct constitutive relationships—provides a more solid foundation for understanding their explanatory success. By eliminating theoretically redundant categories, mechanistic philosophy can focus on developing more precise accounts of the systematic experimental methods, attention to causal complexity, and capacity to integrate evidence across multiple scales that constitute the genuine strengths of mechanistic approaches.

Author contributions Single authorship

Funding N/a

Data Availability N/a

Declarations

Ethical approval N/a

Informed consent N/a

Conflict of Interest N/a

References

Aizawa, K., & Headley, D. B. (2022). Abduction and composition. *Philosophy of Science*, *89*(2), 268–282.

Aizawa, K., & Headley, D. B. (2025). Scientific constitutive abduction. *European Journal for Philosophy of Science*, *15*(2), 1–20.

Armstrong, C. M., & Hille, B. (1972). The inner quaternary ammonium ion receptor in potassium channels of the node of Ranvier. *Journal of General Physiology, 59*(4), 388–400.

Baetu, T. M. (2025). Failures of scale separation in biology and the problem of inter-level causation. *International Studies in the Philosophy of Science, 38*(1), 1–19.

Baumgartner, M., & Gebharter, A. (2016). Constitutive relevance, mutual manipulability, and fat-handedness. *British Journal for the Philosophy of Science, 67*(3), 731–756.

Baumgartner, M., & Casini, L. (2017). An abductive theory of constitution. *Philosophy of Science, 84*(2), 214–233.

Bechtel, W., & Richardson, R. C. (2010). *Discovering complexity: Decomposition and localization as strategies in scientific research*. Princeton University Press.

Bickle, J. (2023). Hodgkin's and Huxley's own assessments of their "quantitative description" of nerve membrane current. *History and Philosophy of the Life Sciences, 45*(3), 1–20.

Catterall, W. A., Goldin, A. L., & Waxman, S. G. (2012). International Union of Pharmacology. XLVII. Nomenclature and structure-function relationships of voltage-gated sodium channels. *Pharmacological Reviews, 57*(4), 397–409.

Catterall, W. A., Raman, I. M., Robinson, H. P. C., Sejnowski, T. J., & Paulsen, O. (2012). The Hodgkin-Huxley heritage: From channels to circuits. *Journal of Neuroscience*, *32*(41), 14064–14073.

Clarke, E., & Jacyna, L. S. (1987). *Nineteenth-Century origins of neuroscientific concepts*. University of California Press.

Craver, C. F., & Bechtel, W. (2007). Top-down causation without top-down causes. *Biology and Philosophy, 22*(4), 547–563.

Craver, C. (2007a). Constitutive explanatory relevance. *Journal of Philosophical Research*, *32*, 3–20.

Craver, C. F. (2007b). Top-down causation without top-down causes. *Biology and Philosophy, 22*(4), 547–563.

Craver, C. F. (2007c). *Explaining the brain*. Oxford University Press.

Craver, C. F. (2009). *Explaining the brain*. Oxford University Press.

Craver, C. F., & Darden, L. (2013). *In search of mechanisms: Discoveries across the life sciences*. University of Chicago Press.

Craver, C. F., Glennan, S., & Povich, M. (2021). Constitutive relevance & mutual manipulability revisited. *Synthese, 199*(3–4), 8807–8828.

Darden, L. (1991). *Theory Change in Science: Strategies from Mendelian Genetics*. Oxford University Press.

Darden, L. (2006). *Reasoning in biological discoveries: Essays on mechanisms, interfield relations, and anomaly resolution*. Cambridge University Press.

DiFrisco, J. (2017). Time scales and levels of organization. *Erkenntnis*, *82*(4), 795–818.

Fagan, M. B. (2015). Collaborative explanation and biological mechanisms. *Studies in History and Philosophy of Science Part A, 52*, 67–78.

Frankenhaeuser, B., & Hodgkin, A. L. (1956). The after-effects of impulses in the Giant Nerve Fibres of Loligo. *Journal of Physiology, 131*(2), 341–376.

Finger, S. (2000). *Minds behind the brain: A history of the pioneers and their discoveries*. Oxford University Press.

Gillett, C. (2020). Why constitutive mechanistic explanation cannot be causal. *American Philosophical Quarterly*, *57*(1), 31–50.

Glennan, S. (2002). Rethinking Mechanistic Explanation. *Proceedings of the Philosophy of Science Association* no. 3 (2002): S342-353.

Glennan, S. (2017). *The New Mechanical Philosophy*. Oxford University Press.

Haüser, S. (2000). The development of the quantal analysis of synaptic transmission. *Progress in Neurobiology*, *62*(4), 427–446.

Hodgkin, A. L., & Huxley, A. F. (1952a). The components of membrane conductance in the giant axon of *Loligo*. *Journal of Physiology, 116*, 473–496.

Hodgkin, A. L., & Huxley, A. F. (1952b). A quantitative description of membrane current and its application to conduction and excitation in nerve. *Journal of Physiology, 117*(4), 500–544.

Hodgkin, A. L., & Huxley, A. F. (1952c). Currents carried by sodium and potassium ions through the membrane of the giant axon of *Loligo*. *Journal of Physiology, 116*, 449–472.

Hodgkin, A. L., & Huxley, A. F. (1952d). The dual effect of membrane potential on sodium conductance in the giant axon of *Loligo*. *Journal of Physiology, 116*, 497–506.

Hodgkin, A. L., Huxley, A. F., & Katz, B. (1952). Measurement of current-voltage relations in the membrane of the giant axon of *Loligo*. *Journal of Physiology, 116*, 424–448.

Kaiser, M. I., & Krickel, B. (2017). The metaphysics of constitutive mechanistic phenomena. *The British Journal for the Philosophy of Science*. https://doi.org/10.1093/bjps/axv058

Kaplan, D. M., & Craver, C. F. (2011). The explanatory force of dynamical and mathematical models in neuroscience: A mechanistic perspective. *Philosophy of Science, 78*(4), 601–627.

Krickel, B. (2018a). *The Mechanical World: The Metaphysical Commitments of the New Mechanistic Approach*. Springer.

Krickel, B. (2018b). Saving the mutual manipulability account of constitutive relevance. *Studies in History and Philosophy of Science, Part A, 68*, 58–67.

Levy, A. (2013). What was Hodgkin and Huxley's achievement? *British Journal for the Philosophy of Science, 65*(3), 469–492.

Machamer, P., Darden, L., & Craver, C. F. (2000). Thinking about mechanisms. *Philosophy of Science, 67*(1), 1–25.

McComas, A. J. (2011). *Galvani's spark: The story of the nerve impulse*. Oxford University Press.

Oppenheim, P., & Putnam, H. (1958). Unity of science as a working hypothesis. *Minnesota Studies in the Philosophy of Science*, *2*, 3–36.

Potochnik, A. (2021). Our World Isn't Organized into Levels. In Daniel Stephen Brooks, James DiFrisco & William C. Wimsatt, *Levels of Organization in the Biological Sciences*. Cambridge, Massachusetts: MIT Press.

Potochnik, A., & McGill, B. (2012). The limitations of hierarchical organization. *Philosophy of Science*, *79*(1), 120–140.

Raman, I. M., & Ferster, D. L. (2022). *The annotated Hodgkin and Huxley: A reader's guide*. Princeton University Press.

Romero, F. (2015). Why there isn't inter-level causation in mechanisms. *Synthese, 192*(11), 3731–3755.

Ross, L. N. (2021). Causal concepts in biology: How pathways differ from mechanisms and why it matters. *British Journal for the Philosophy of Science, 72*(1), 131–158.

Ross, L. N., & Bassett, D. (2024). Causation in neuroscience: Keeping mechanism meaningful. *Nature Reviews Neuroscience, 25*, 81–90.

Simon, H. A. (1962). The architecture of complexity. *Proceedings of the American Philosophical Society, 106*(6), 467–482.

Wimsatt, W. C. (2007). *Re-engineering philosophy for limited beings: Piecewise approximations to reality*. Harvard University Press.

Woodward, J. (2003). *Making things happen: A theory of causal explanation*. Oxford University Press.